

ESP32 IoT Weather Station Project Report

This project presents an IoT-based weather monitoring system using the ESP32 microcontroller, DHT22 temperature and humidity sensor, and OLED display. The system continuously monitors environmental conditions and displays the measured values in real time.

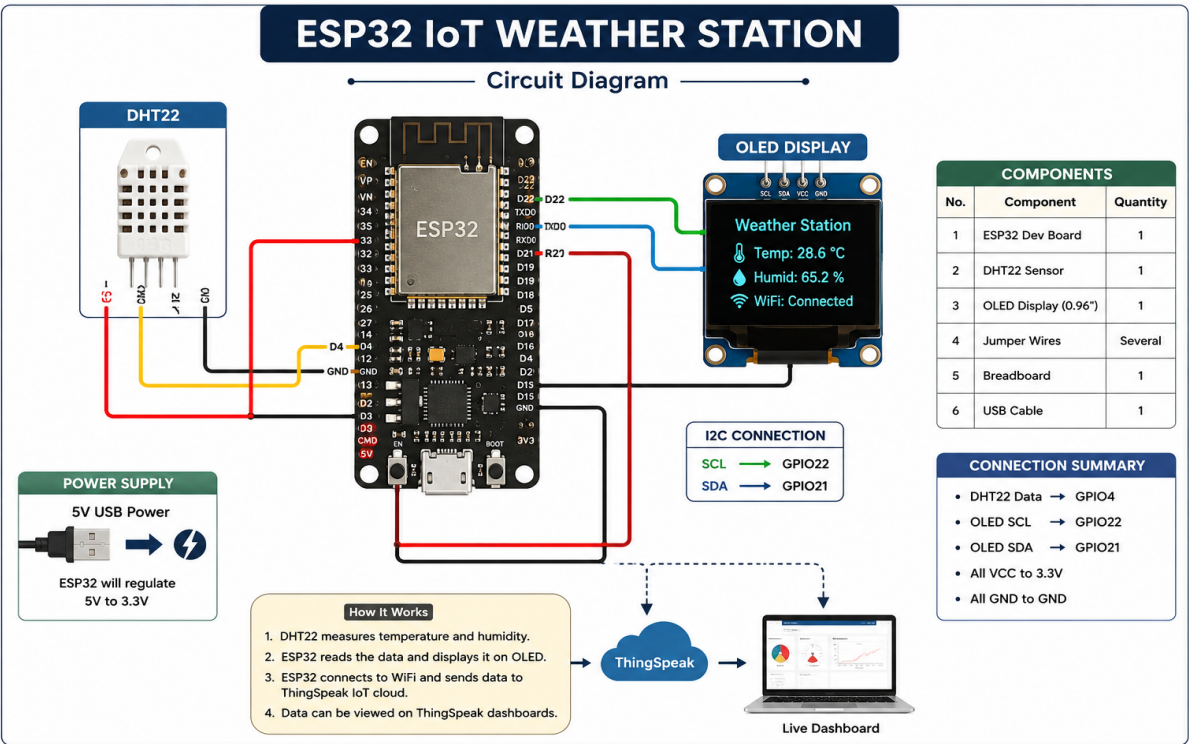
Objectives
Monitor temperature and humidity in real time
Display sensor readings on OLED screen
Enable IoT-based embedded system learning
Provide a simulation-ready Wokwi project

Component	Quantity
ESP32 DevKit V1	1
DHT22 Sensor	1
SSD1306 OLED Display	1
Jumper Wires	Several

Working Principle:

The DHT22 sensor measures temperature and humidity from the surrounding environment. The ESP32 reads the sensor values and processes the data. The processed values are then displayed on the OLED display and simultaneously printed to the serial monitor. The ESP32 can also connect to Wi-Fi networks for cloud-based IoT applications.

Circuit Diagram



Advantages:

- Low-cost embedded IoT solution
- Real-time monitoring capability
- Easy to simulate using Wokwi
- Compact and energy efficient
- Suitable for academic mini projects

Applications:

- Smart homes
- Weather monitoring stations
- Greenhouse monitoring
- IoT learning projects
- Environmental data collection

Conclusion:

The ESP32 IoT Weather Station demonstrates how embedded systems and IoT technologies can be integrated for environmental monitoring. This project provides hands-on experience with sensors, microcontrollers, displays, and wireless communication technologies.